

AAYANSH YADAV

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Yadav-Aayansh Yadav-Aayansh



IIT Madras
B.S. Degree

Summary

Software Engineer specializing in backend development with expertise in FastAPI, Spring Boot, Flask, PostgreSQL, and Redis. Experienced in full-stack application development using Vue.js and Nuxt.js, delivering scalable and production-ready solutions.

Education

Indian Institute of Technology, Madras

Bachelor of Science in Data Science and Applications

Chennai, Tamil Nadu, India

2023 – 2027 (Expected)

- **CGPA:** 8.81/10
- **Relevant Courses:** Mathematics, Statistics, Database Management Systems, Machine Learning, Data Structures & Algorithms

Experience

Backend Intern

Aug 2025 – Present

Straive

- Developed and deployed **CyberDetect**, a browser-based security log analysis platform featuring AI-powered threat detection, real-time dashboards, and client-side data privacy.
- Built **Caption Generator**, an AI-driven subtitle generation tool using Google Gemini + OpenAI pipelines for multimodal analysis, automated compression, and professional SRT export.
- Delivered multiple internal projects leveraging FastAPI, Python, and JavaScript, improving backend efficiency and enabling scalable workflows.
- Technologies: FastAPI, Python, JavaScript, React, Tailwind, OpenAI/Gemini APIs

Software Developer Intern

May 2025 – July 2025

Annam.AI

- Built a modular FastAPI backend with 5+ services (auth, job board, chat, rentals), tested under 500+ concurrent users using Locust.
- Implemented OTP verification with Redis and JWT, reducing session errors by 35% during load testing.
- Integrated Uber H3 geospatial indexing, reducing job search latency from 350ms to 70ms for 1–10km queries.
- Technologies: FastAPI, PostgreSQL, Redis, Firebase, JWT, Uber H3

Projects

HomeMate: Household Services

GitHub Repository

- Developed a multi-user platform connecting customers with verified service professionals, managed by an admin.
- Integrated Redis caching and Celery for async task management, reducing API response times by 30–40%.
- Implemented JWT-based authentication and strict role-based permissions to secure user sessions.
- Tech Stack: Flask, Vue.js, JWT, HTML, CSS, Bootstrap, Chart.js, Redis

NoctiWave: Influencer Engagement & Coordination Platform

GitHub Repository

- Developed a web application enabling sponsors to manage campaigns and influencers to handle ad requests.
- Tech Stack: Flask, HTML, CSS, Bootstrap, Chart.js

Technical Skills

- **Languages:** Python, Java, SQL, JavaScript
- **Backend:** FastAPI, Flask, Spring Boot, PostgreSQL, Redis
- **Frontend:** Vue.js, Nuxt.js, HTML, CSS, Bootstrap, Chart.js
- **Machine Learning:** Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn
- **Tools & DevOps:** Git, Docker, Linux, AWS, GCP

Achievements

- Kaggle Competition Winner: Ranked #1 out of 1,727 entrants for System Threat Forecaster Model.
- Machine Learning using NumPy Workshop: Hands-on ML training with NumPy-based implementations. Certificate